

Opponent's assessment of
'Enumerating score sequences and permutations
by inversions and forbidden patterns'
submitted by
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1 Assessment of the thesis

The thesis comprises four papers (the first three of which have already been published, the last is submitted), plus an introduction. There is certainly enough material here for a PhD, and indeed some of the mathematics contained in the thesis is both elegant and important. Thus, I have no hesitation in recommending it for public defence.

The writing style is generally excellent. There is a slight mismatch in style between the first two papers and the last two, presumably as a result of the fact that papers I and II have several authors, whereas papers III and IV are authored by the candidate alone. For the most part, I found very few typos or mistakes, although there are some issues in paper IV (detailed below).

I will make some more specific comments, paper-by-paper, in this section, and thereafter there are two further sections: Section 2 lists some questions for the candidate to consider (some of these questions will form the basis for discussion during the defence). Section 3 gives a longer list of minor issues and typos for the candidate to consider in preparing a revised version of the thesis.

Paper I, published by *Proceedings of the AMS* in 2023, concerns the enumeration of 'score sequences' of tournaments, and is rather different from the rest of the thesis (though there are some connections with paper II: see later). The published version of this very attractive paper seems to have gathered a considerable amount of attention since it was published, and I would like to explore some of these consequences during the defence.

Paper II, published in the *Electronic Journal of Combinatorics* in 2023, commences the study in the thesis of permutations, and specifically considers the problem of enumeration permutations of length n with exactly $n-i$ inversions. Writing $S_i(x)$ to denote the generating function that counts these permutations, the paper establishes two main results. The second of these results concerns an expression for $S_0(x) = R(xC(x))$, where $R(x)$ is the generating function for a somewhat complicated-looking signed family of partition-like objects. The

first proceeds via an elegant argument that factors each such permutation into two parts, and shows that $S_{i+1}(x) = xC(x)S_i(x)$, where $C(x)$ is the generating function for the Catalan numbers. This formula bears a striking resemblance to the formula

$$S^\circ(x) = xC(x)S(x)$$

established in Corollary 11 of paper I, and I wondered whether anything beyond the superficial similarity is true? For example, given a generating function $f(x)$ that counts some family of objects, what does $xC(x)f(x)$ count? How common is this phenomenon?

Paper III, published in the special issue of *Discrete Mathematics and Theoretical Computer Science* for Permutation Patterns 2024, is a single-author contribution which considers the task of counting permutations not by their length, but by the number of inversions they contain. Unsurprisingly perhaps, this seems to be quite hard to do, but the candidate makes some progress in permutation classes avoiding one or more patterns of length three. There are some fairly exotic bijections, to partitions and fountains of coins (among others), and this enables progress to be made. Some of the later classes (that avoid two or more patterns of length three) have very clearly defined structure, and while the candidate's proofs demonstrate the application of their earlier bijections to these restricted scenarios, in some cases there are rather more direct, elementary proofs available. I mention some of these in the next section.

Paper IV, submitted in 2025 for publication, is primarily motivated by the search for improved bounds on the growth rate of $\text{Av}(1324)$. Subject to a conjecture concerning walks in the 'grouped 1324-avoiding graph', this work would imply a new lower bound of 10.418 for $\text{gr}(\text{Av}(1324))$, which would be impressive given the level of effort put into establishing the existing bounds.

There are a few issues with this paper as currently written, though these issues would appear to be superficial (that is, I believe the underlying mathematics to be largely correct), and I invite the candidate to address these carefully before the defence. Perhaps a more significant criticism, however, is that the notion of building permutations by iteratively adding new maximal elements has quite a long and varied history, and this does not seem to be well-represented at present.

2 Questions for the candidate

Notation: (m, n) means 'page m , line n '. If $n < 0$, then count up from the bottom of the page.

Paper I: Can you update us on what has happened since this paper was published? I see it has various citations, so it would be good to explore the impact of this paper to-date.

(24,17) This decomposition of a tournament is part of a more general theory, that applies to tournaments, but also to all other kinds of combinatorial structures, called the modular decomposition (or 'substitution decomposition'). What can you say about this? e.g. notable references?

Paper II: I'd like to learn more about the motivations for the work in this paper. For example, what are advantages of the subdiagonal generating function expressions that you find, over and above the expressions that we can derive from the work of Knuth–Netto?

The Mahonian numbers are, rightly, described as ‘famous’. What are some interesting places that they appear?

(42,6) There are obvious parallels between Theorem 3, and the expression found in Paper I. This is surely more than a coincidence – have you investigated it at all?

Paper III:

(63,14) and (63,19) ‘Experimental results’ are mentioned twice here, but it would not be hard to get these results properly. This could be done either directly, or by appeal to the literature. How would you approach getting precise answers here?

(64,–13) The ‘inversion table’ is defined the opposite way in this paper to how it is in Paper II. Do these two viewpoints coexist throughout the topic literature?

(70,10) ‘computing new terms’ Did you do this? How does it compare with the various continued fractions that can alternatively be used to define A006958?

Pages 79–81: There are several results here where the classes in question have particularly simple structure:

Theorem 16: $\text{Av}(132, 213)$ are just the permutations made up of skew sums of increasing sequences. Do such structural observations allow for any intuition into the (somewhat complicated-looking) result here?

Theorem 18: The permutations in $\text{Av}(132, 321)$ are ‘V-shaped’. In this case, I think this may give another way to derive Theorem 18 – have you tried this?

Theorem 20: There is definitely an elementary proof of this result, can you find it?

Theorem 21: I’m not sure the structure of the permutations helps much here, but have you tried?

Paper IV:

(90,–7) This directed graph has been studied for quite a long time, under the name ‘Catalan tree’, but is actually part of a broader study of ‘generating trees’. It would be nice to see some of this historical context here, and indeed some comparison with other procedures which have built permutations iteratively by adding a new maximal element.

(97,–9) To what extent is it feasible to make significant memory/computation improvements to go further? The number of states is growing exponentially, so even a single extra term in the sequence is going to take a while, and Figure 6 suggests one more term will make much of a difference.

The ideas in this paper, and the potential improved lower bound, are interesting ideas. The existing record lower bound of 10.271 is formed from a very precise structural characterisation based on stitching together ‘dominoes’ (whose growth rate is known precisely). Could a hybrid approach be developed, utilising what we know about the domino to inform an insertion mechanism such as you have here?

3 Minor Corrections/typos

Notation: (m, n) means ‘page m , line n ’. If $n < 0$, then count up from the bottom of the page.

Cover page: ‘Brignal’ $\mapsto$ ‘Brignall’ please!

(6,−5) Figure 5: It might make more sense for trains to enter the stack from the right, given how you describe the process in which the leftmost entry of a permutation enters the stack first?

(11,−12) You reference Arratia’s paper [2] here, but his name should be mentioned too: Marcus and Tardos only establish existence of the limsup. A similar comment applies on page 89.

(12,6) Is there a reference for Hanna’s conjecture (e.g. give the OEIS sequence)?

(22,16) ‘Viewing i is’ $\mapsto$ ‘Viewing i as’

(23,9) Missing cross-reference.

(23,−2) and (26,16) $\oplus$ is being used here on tournaments. This is not defined (but the meaning is clear). Perhaps warn readers about this slight abuse of notation? Similar comment applies to writing ‘ $S + |V|$ ’.

(42,4) What does $\simeq$ mean here?

(43,−9) and (43,−5) ‘counts’ doesn’t feel like quite the right word here, because of the signed nature of the objects. I personally would be happier with terms like ‘signed sum’, e.g. ‘the coefficient of x^n is equal to the signed sum of ...’

(43,−3) The re-use of $\vdash$ for this new object confused me for a while. I was going to suggest you use $\Vdash$ instead, but that appears later, so maybe just warn the reader that this is not being treated just as an integer partition.

(63,5) You use the term ‘growth rate’ here (which is just a number), but you’re actually talking about more precise first-order asymptotics.

(64,3) to (64,4) ‘In Section 3 ... previous maps.’ is not currently a sentence.

(65,13) ‘we must have $\tau = \pi \oplus (1 \oplus \cdots \oplus 1)$ ’ might merit more explanation. For example, a statement like ‘adding a trailing zero to the inversion table of π results in the inversion table of $\pi \oplus 1$ ’, possibly with some words of explanation.

(68,−10) ‘after adding ...trailing zeroes’ Feels like something has been missed here, and perhaps it would be better to discuss appending zeroes to a sequence of integers before stating the lemma?

(71,Figure 3) Somewhere on p70, explain the convention about labelling the entries as ‘0’ when a red coin on the bottom level was deleted before we reached that entry. (Once the coin no longer exists, there is nothing to label according to the current procedure.) This might then save the slightly awkward explanations at the top of p71.

(72,−7) You could submit the sequence arising from $I_k(123)$ to OEIS.

(75,2) ‘removing no coins counts as a rectangle’ I think this would be better defined before stating Theorem 8.

(75,−4) I don’t think you can ‘enumerate’ Pascal’s triangle. What you mean is that $|I_k(132, 123)|$ is equal to the k th entry of OEIS sequence A135278, which is equal to the en-

tries of Pascal's triangle, reading row-by-row, from left-to-right, after removing the leftmost column.

(90,4) 'taht' $\mapsto$ 'that'

(94,-14) 'It suffices to consider' in what way does it 'suffice'? What you mean needs to be stated more precisely: it has something to do with the quotient graph giving a valid lower bound on the growth rate.

(96,-4) Should 'Av(213)' be 'Av(312)' here?

(96,-3) The discussion in the preceding pages has been quite informal. This is fine, but I don't think you can do this while also claiming to prove something by induction. So, at this point, more precision is needed: which of the statements in the preceding paragraph is being proved? What is the induction on? (Maybe the length of the walks?) Similar comments apply over to the top of p97 too.

p97 Definition 3: There are a few things:

- 'whose last element is fixed' might more usually be 'that ends with its maximum'

- Does the avoider graph of π *really* comprise all elements of $\text{Av}(\pi)$ up to some length? I expected you to form something like π' by removing the last (maximal) element, and build nodes labelled by entries from $\text{Av}(\pi')$. [Alternatively, maybe you're building the avoider graph of $\pi \oplus 1$?]

(98,2) In $W_{n,k}(\pi)$, what is n ?

(98,20) 'The value $|\pi|$ always appears...' Should this be $|\pi| + 1$? In that case, however, your definition of S' will have some entries equal to -1 , rather than the zero that you're relying on. Something's not quite right in what's written here.

(98,21) 'so we can always remove one zero' Clearer, perhaps, to describe how inserting to the left of all entries will always yield the same, and so we ignore it, thus focussing solely on the possible insertions that come after at least one element of pi ?

(98,-10) $|\tau| + |\pi_R|$. This does not look right to me, since this is the contribution to S' , not S . Perhaps instead of $|\tau|$ it should be $|\pi_L| - |\tau|$, but that may not account for the off-by-1 issues introduced earlier (and that statement would simplify).

(98,-4) Similar issue here: I don't think $|\tau|$ is the contribution to S' in this case.

(102,-4) to remove no elements ... maxima.' This feels like an observation that might have been useful earlier on? For example, if $|\pi| = n$, then the number of times we see $n + 1$ in S (or 0 in S') when inserting into π is equal to the number of right-to-left maxima (plus 1 for insertion before all points).

(104,8) 'constant time' Seems that the time to compute $E(n, r, m, s)$ is generally going to depend on n and r , at least. Maybe you just mean subexponential?

(104,-1) Be careful talking about the 'growth rate of walks'. This is not defined, but it may not even exist. Probably the best you can do is $\limsup$.

(105,4) 'Though the above conjecture ... the case $n = k$.' is not a sentence.

(105,-1) Clarify what you mean by 'converge speed'.